

Linear Circuit Analysis

(EE1001)

Date: April 3rd 2024

Course Instructor(s)

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|------------------------|------------------|
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Sessional-II Exam

Total Time (Hrs):	1
Total Marks:	40
Total Questions:	2

Roll No

Section

Student Signature

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1. Attempt all the questions.
2. Attempt all parts of the same question together.
3. Show all the steps with proper circuit diagrams and answers with proper units.

CLO # 02: Apply DC circuit analysis techniques and theorems to solve linear resistive circuits.

Q1: For the circuit shown in Figure 1 below,

[20 marks]

- (a) Calculate the Thévenin Equivalent voltage, V_{Th} with respect to the terminals a,b. [8]
- (b) Calculate the Thévenin Equivalent resistance, R_{Th} at a,b. [8]
- (c) Sketch the Thévenin and Norton Equivalent circuits. [4]

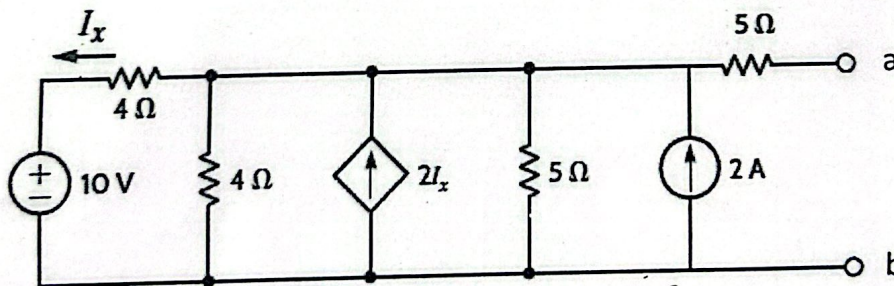


Figure 1. Circuit for Thévenin Equivalent

CLO # 03: Solve circuits containing ideal operational amplifiers

- Q2:** For the circuit shown in the Figure 2 below, the op amp is ideal. [20 marks]
- (a) Label the five op amp terminals with their names. (Draw a separate diagram for the circuit symbol and then label it). [5]
- (b) Express the output voltage v_o in terms of the input voltage sources; v_a and v_b by assuming linear region of operation. [10]
- (c) If $v_b = 2\text{ V}$, determine the range of v_a such that the amplifier does not saturate. [5]

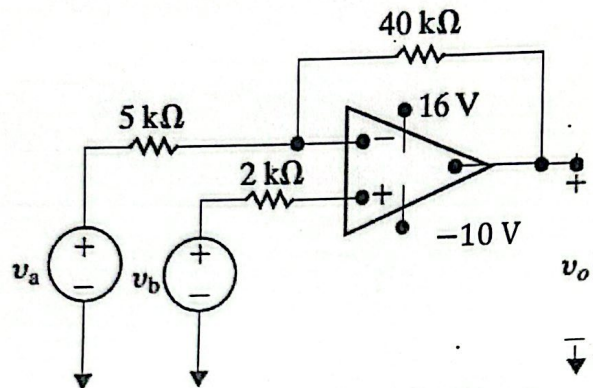


Figure 2. An Operational Amplifier Circuit